ReHQJ: the HexOpt Paper and `conspire`

autotwin/automesh issue #768

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Sources

This note compares the HexOpt objective in two sources.

- **Paper.** H. Tong and Y. J. Zhang, "Fast and Robust Hexahedral Mesh Optimization via Augmented Lagrangian, L-BFGS, and Line Search," arXiv:2410.11656, Equations (2.5), (2.6), (2.8), and (2.9).
- `conspire`. Branch `msj` at commit `515c78cee`, draft mrbuche/conspire.rs#251, file `src/geometry/mesh/quality/improve/msj/mod.rs`, functions `evaluate` and `terms`.

h is one hex element of the mesh H, and $\ell(h)$ is its average edge length. $J(h)$ is the minimum of the Jacobian $\det(\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2)$ over nine points of h: its eight corners and its body center (paper, Section 2.1). At a corner, $\mathbf{e}_0$, $\mathbf{e}_1$, and $\mathbf{e}_2$ are the three edge vectors that meet there. At the center, they join the centers of opposite faces. $\hat{J}(h)$ is the minimum of the scaled Jacobian $\det(\hat{\mathbf{e}}_0, \hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2)$ over the same nine points, where $\hat{\mathbf{e}}_k = \mathbf{e}_k/|\mathbf{e}_k|$. The two minimums can occur at different points, so $\hat{J}(h)$ is in general not $J(h)$ divided by a product of lengths. `conspire` computes both minimums in `terms`, each over all nine points. $\Theta \in [0, 1]$ is the threshold for the minimum scaled Jacobian $\hat{J}$. Both sources hold ℓ fixed when they take the gradient.

This note's notation differs from the paper's in four symbols. Subscripts p and c mark the paper's and `conspire`'s forms of ψ.

quantity	paper	this note	remark
per-hex term	$ReHQJ(h, \Theta)$	$\psi(h)$	consistent with ψ, the Helmholtz free energy density in continuum mechanics
scaled Jacobian	$SJ(h)$	$\hat{J}(h)$	shorter, derives from normalization $\hat{(\cdot)}$
average edge length	$\bar{e}$	ℓ	$\bar{e}^2$ can be confused with e^{-2}
augmented Lagrangian	L	$\mathcal{L}$	conventional symbol for a Lagrangian

Definitions

Both sources use the same term for an inverted hex:

$$\psi_{\mathrm{p}}(h) = \psi_{\mathrm{c}}(h) = J(h)/\ell, \qquad J(h) \leq 0. \tag{1}$$

For $J(h) > 0$, the paper defines

$$\psi_{\mathrm{p}}(h) = \begin{cases} \ell^2\, \hat{J}(h), & \hat{J}(h) \leq \Theta, \\ \Theta, & \hat{J}(h) > \Theta, \end{cases} \tag{2}$$

and `conspire` implements

$$\psi_c(h) = \ell^2 \min\big(\hat{J}(h), \Theta\big). \tag{3}$$

The two differ only above Θ, where $\psi_p = \Theta$ and $\psi_c = \Theta\,\ell^2$. ψ_c is one expression: ℓ^2 times $\hat{J}$ clamped at Θ. ψ_p has no such form unless $\ell = 1$. Figure 1 plots both forms.

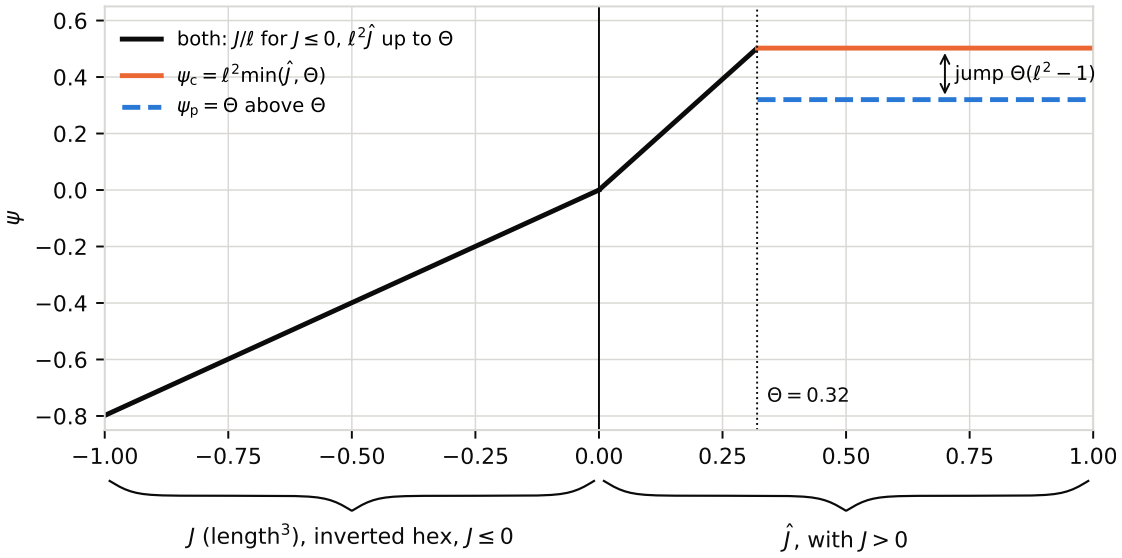


Figure 1: The per-hex term ψ, with $\ell = 1.253$, the median on the snapped bone octree, and $\Theta = 0.32$. The braces mark the two domains. For an inverted hex, $J \le 0$, both forms are J/ℓ, plotted against J. For $J > 0$, both forms follow $\ell^2 \hat{J}$ up to Θ, plotted against $\hat{J}$. Above Θ, ψ_c stays at $\Theta\,\ell^2 = 0.502$, and ψ_p drops to $\Theta = 0.32$.

Units

J has units of length^3, since each $\mathbf{e}_k$ has units of length. $\hat{J}$ and Θ have no units.

case	ψ_p	ψ_c
$J \le 0$	length^2	length^2
$J > 0$, $\hat{J} \le \Theta$	length^2	length^2
$\hat{J} > \Theta$	**none**	length^2

The minimized function

At each threshold Θ, `conspire` minimizes the augmented Lagrangian

$$\mathcal{L} = -\sum_{h \in H} \psi_c(h) + \sum_i \Big[\lambda_i\, r_i + \frac{\rho}{2}\, r_i^2\Big]. \tag{4}$$

- The first sum has one term per hex, with ψ_c as defined above.

- The second sum holds the boundary on the target STL, with a multiplier term and a quadratic penalty. $r_i = (\mathbf{n}_i \cdot \mathbf{x}) - d_i$ is the signed distance from a boundary node $\mathbf{x}$ to the plane of one STL triangle. $\mathbf{n}_i$ is the triangle's unit normal, and $d_i = \mathbf{n}_i \cdot \mathbf{y}$ for any vertex $\mathbf{y}$ of that triangle. λ_i is the Lagrange multiplier for plane i, and ρ is the penalty coefficient. Each boundary node gets up to four planes: the plane of its closest triangle, plus the planes of nearby triangles whose normals differ from those already kept by more than 30°.
- After each L-BFGS run, `conspire` updates each multiplier, $\lambda_i \leftarrow \lambda_i + \rho\, r_i$. ρ starts at 100 and doubles each time the mesh meets Θ with the boundary on the STL.
- r_i and λ_i have units of length, and ρ has none. Both constraint terms have units of length^2.

The paper's Equation (2.8) has the same structure. Its constraint is $\mathbf{Z}_k - \mathbf{Z}_k^t$, the offset from each boundary vertex to its closest point on the target surface. Its Equation (2.9) updates the multipliers the same way, and it doubles ρ each time the minimum scaled Jacobian reaches Θ. The paper's L uses ψ_p in place of ψ_c. The next section returns to that difference.

Behavior at the threshold

1. **Continuity.** At $\hat{J} = \Theta$, both forms below Θ give $\Theta\,\ell^2$. Above Θ, ψ_c gives the same value, so it is continuous at Θ. Above Θ, ψ_p gives Θ, so it jumps by $\Theta(\ell^2 - 1)$ unless $\ell = 1$. Neither form is differentiable at Θ. The slope drops to zero there in both, by design.
2. **The sign of the jump.** $\mathcal{L}$ in Equation (4) sums $-\psi$ over all hexes. Under ψ_p, a hex that rises above Θ changes $\mathcal{L}$ by $+\Theta(\ell^2 - 1)$. For $\ell > 1$, $\mathcal{L}$ rises. The Armijo condition requires $\mathcal{L}$ to fall, so the jump counts against every step that lifts such a hex above Θ. For $\ell < 1$, $\mathcal{L}$ falls, so the jump rewards the crossing instead. The sign depends on the length unit of the mesh.
   - On the snapped bone octree in the reference frame, ℓ runs from 0.644 to 2.81, with a median of 1.253. The same mesh has hexes on both sides of $\ell = 1$.
   - On the scale-5 bone mesh in the STL's own units, ℓ runs from 0.009 to 0.034.

   Under ψ_c, the crossing changes $\mathcal{L}$ by zero.
3. **Units.** ψ_c has units of length^2 in every case. `conspire`'s types enforce this: each term is a `Quantity<Area>`.

For a hex that stays above Θ through a step, both forms are constants. They give the same gradient and the same energy differences. The forms differ only when a hex crosses Θ.